Does Air Quality Explain Adult Asthma? A County-Level Analysis of CDC and EPA Data

Abstract

Air pollution is a well-known asthma trigger, yet it is unclear whether county-level air quality predicts how common asthma is across the United States. This report asks whether median Air Quality Index (AQI) is associated with adult asthma prevalence across U.S. counties, with attention to what the pattern suggests for the Los Angeles area. I merged CDC PLACES county estimates of adult asthma prevalence (2025 release) with EPA Annual AQI by County data (2026), which gave 769 counties with both measures. I used correlation, group comparisons (Welch's t-test, Mann-Whitney U, ANOVA, chi-square), and regression with controls for county population and state. Air quality showed no meaningful relationship with asthma prevalence. The correlation was -0.04, counties in the lowest and highest AQI quartiles averaged 10.67% and 10.61%, and median AQI was not a significant predictor in either regression ($p = 0.19$ and $p = 0.48$). County population and state explained far more of the variation. These results reflect the limits of county averages, adult-only estimates, and missing socioeconomic data, not proof that air pollution is harmless. Future work should use census-tract data, income measures, and child asthma data.

1. Introduction

Asthma affects millions of adults in the United States, and air pollution is one of the most widely studied triggers. Fine particles (PM2.5), ozone, and nitrogen dioxide can irritate the airways and worsen symptoms. If cleaner air means less asthma, counties with better air quality should show lower asthma prevalence.

The Los Angeles area is a natural place to ask this question. It has long faced air-quality challenges, and Los Angeles County public health research reports that asthma outcomes vary across neighborhoods and with income and race.

This report asks: **Is county-level air quality (median AQI) associated with adult asthma prevalence across U.S. counties, and what does the pattern suggest for the Los Angeles area?** It combines two public datasets, CDC PLACES asthma estimates and EPA air quality summaries, covering 769 counties. Section 2 describes the data and methods, Section 3 presents the findings, and Sections 4 and 5 discuss limitations and next steps.

2. Data and Methodology

2.1 Data sources

- **Asthma prevalence:** CDC PLACES, County Data (GIS-friendly format), 2025 release. It gives model-based estimates of current asthma among adults aged 18 and older for 3,143 counties, built from the Behavioral Risk Factor Surveillance System (BRFSS) survey, Census population estimates, and American Community Survey data.

- **Air quality:** EPA Annual AQI by County, 2026. It has one row for each of 827 monitored counties, with the median AQI, counts of unhealthy days, and the number of days each pollutant (CO, NO2, ozone, PM2.5, PM10) determined the AQI.

2.2 Data preparation

- I standardized state names, trimmed extra spaces, and merged the files on state and county name.
- I excluded counties missing either value instead of filling them in. In PLACES, 187 counties have no asthma estimate. Of the 827 AQI records, 27 matched no PLACES county because of naming differences, Alaska boroughs, Puerto Rico, and one Mexico record, and 34 matched counties had no asthma estimate. That left **769 counties**.
- I derived four variables: the share of days rated "Unhealthy for Sensitive Groups" or worse, the dominant pollutant (the one with the most days), AQI quartile tiers, and the log of county population, because population is heavily right-skewed.
- Three counties (Baltimore, Fairfax, and Roanoke) matched both an independent city and a same-named county, so each appears twice. Removing them leaves the results essentially unchanged.

2.3 Statistical methods

- **Descriptive statistics and correlation:** means, medians, and Pearson correlations between asthma prevalence and AQI measures.
- **Group comparisons:** a Welch's t-test comparing the lowest and highest AQI quartiles, with a Shapiro-Wilk normality check and a Mann-Whitney U test as a non-parametric backup. A one-way ANOVA compared asthma across dominant-pollutant groups, and a chi-square test compared AQI quartile with above-median asthma.
- **Regression:** ordinary least squares predicting age-adjusted asthma prevalence from median AQI, controlling for log population. A second model added state fixed effects with standard errors clustered by state.
- Correlations and group comparisons use crude prevalence. Results are significant at $p < 0.05$.

All analysis was done in Python in Google Colab, using Pandas, NumPy, SciPy, statsmodels, Matplotlib, and Seaborn.

3. Key Findings

3.1 Counties look alike on both measures

Adult asthma prevalence is roughly bell-shaped, averaging 10.6% with a range of 8.0% to 14.3%. Median AQI centers near 41, with a tail of cleaner counties. Few counties stand out on either measure, so large differences are unlikely to appear.

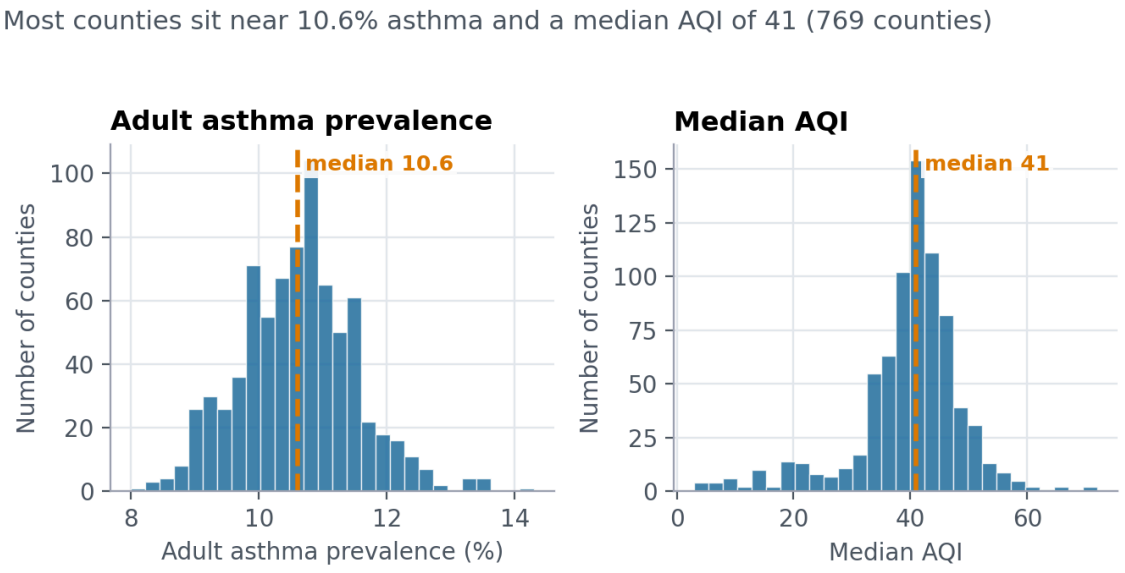


Figure 1. Distribution of adult asthma prevalence (left) and median AQI (right) across 769 U.S. counties. Dashed lines mark the medians (10.6% and 41).

3.2 Worse air quality does not go with higher asthma

The correlation between median AQI and asthma prevalence is essentially zero (r = -0.04). The share of unhealthy days (r = -0.11), PM2.5 days (r = 0.00), and ozone days (r = -0.03) tell the same story.

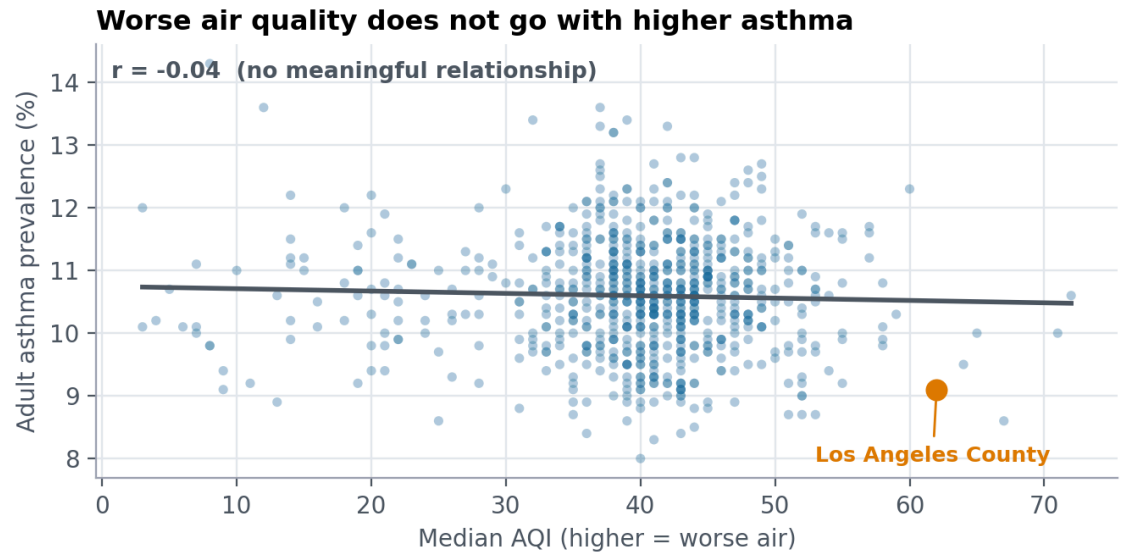


Figure 2. Adult asthma prevalence versus median AQI for each county. The trend line is nearly flat (r = -0.04). Los Angeles County is highlighted.

Counties in the lowest and highest AQI quartiles averaged 10.67% and 10.61%. The Shapiro-Wilk test showed mild non-normality in the lowest group (p = 0.004), so I confirmed the result with a Mann-Whitney U test.

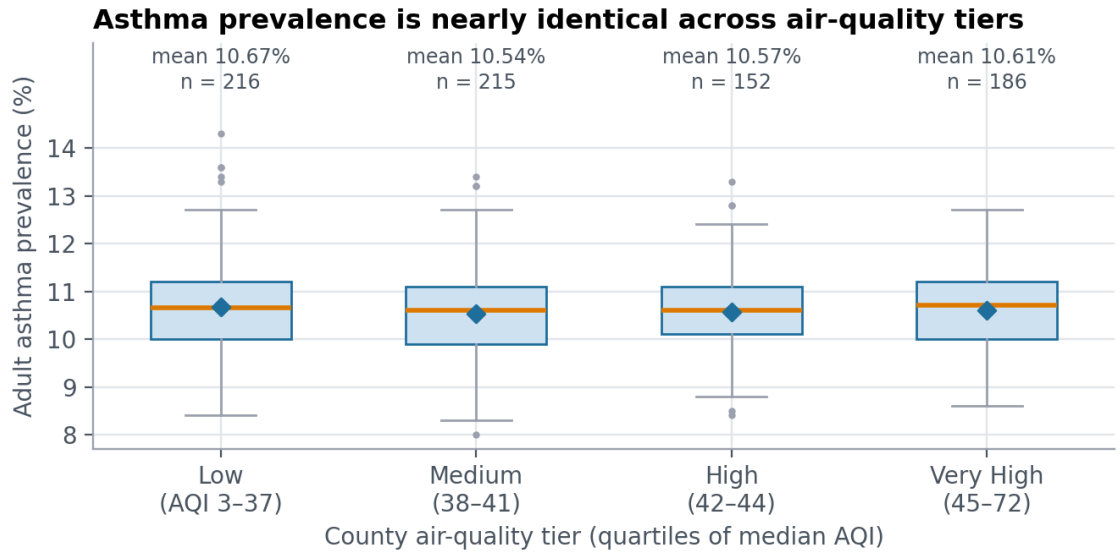


Figure 3. Adult asthma prevalence by AQI quartile. Boxes show the middle 50% of counties, orange lines the medians, and diamonds the means.

Every test agrees that there is no meaningful difference (Table 1).

Table 1. Summary of statistical tests of air quality and adult asthma prevalence (769 counties).

Test	Comparison	Result	Interpretation
Pearson correlation	Median AQI vs. asthma prevalence	r = -0.04	No meaningful relationship
Welch's t-test	Lowest vs. highest AQI quartile	t = 0.70, p = 0.48	No difference (10.67% vs. 10.61%)
Mann-Whitney U	Same groups, non-parametric	U = 20,348.5, p = 0.82	No difference
One-way ANOVA	Asthma by dominant pollutant	F = 1.09, p = 0.35	No difference
Chi-square	AQI quartile vs. above-median asthma	χ^2 = 3.16, p = 0.37	No association
OLS regression	Adjusted asthma ~ median AQI + log population	AQI: 0.005 per point, p = 0.19; R^2 = 0.09	AQI not significant
OLS with state effects	Same model plus state	AQI: 0.004, p = 0.48; R^2 = 0.67	AQI not significant

3.3 County size and state matter more than air quality

Larger counties report slightly lower asthma prevalence (r = -0.26), and log population is the only strong predictor in the regression ($p < 0.001$). Asthma also varies by state: the highest county averages appear in Maine, Oregon, West Virginia, Massachusetts, and Oklahoma. Adding state effects raises the share of variation explained from 9% to 67%, while AQI stays non-significant.

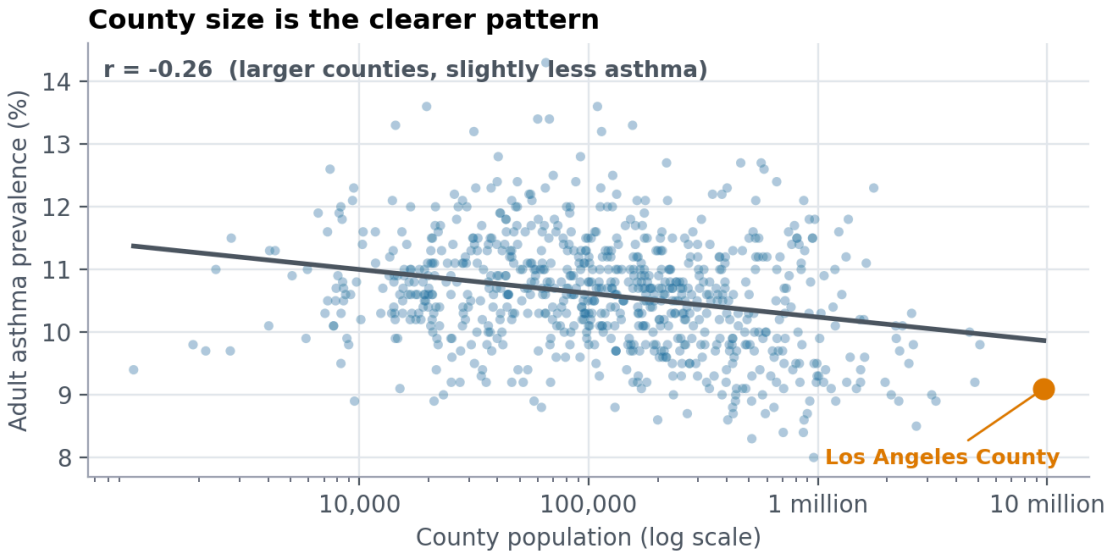


Figure 4. Adult asthma prevalence versus county population (log scale). Larger counties report slightly lower prevalence (r = -0.26).

3.4 The Los Angeles area

Among the 40 California counties in the sample, Los Angeles County has one of the highest AQI values (62) but below-average asthma prevalence (9.1%, compared with 10.6% overall). Orange (9.0%) and San Diego (8.9%) look similar. The highest asthma appears in northern counties such as Trinity and Humboldt (11.0%) with moderate AQI.

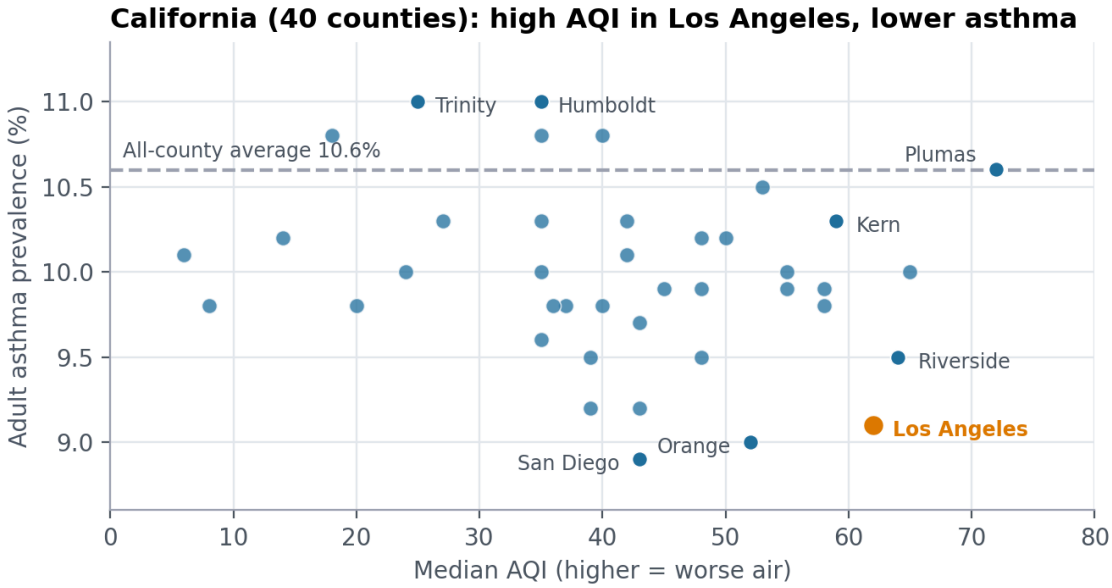


Figure 5. Adult asthma prevalence versus median AQI for 40 California counties. Los Angeles County (orange) has high AQI but below-average asthma.

4. Discussion

4.1 What the results mean

In this data, counties with worse air quality do not have higher adult asthma prevalence. Six different methods agree, so the null result is not an artifact of one test. Where a county is, and how large it is, say more about its asthma level than its AQI does.

Some counties break the expected pattern. Apache and Navajo counties in Arizona have among the highest asthma prevalence (14.3% and 13.6%) at very low AQI (8 and 12), while large metro counties such as Los Angeles, Orange, and Queens have low asthma at high AQI. County air quality alone would therefore be a poor guide for targeting asthma programs.

4.2 Why county data may hide an effect

- **Averaging:** a county median hides neighborhood exposure. A study of Los Angeles County found that asthma emergency visits cluster geographically, with elevated rates north of the major ports, even after adjusting for socioeconomic and environmental factors.
- **What AQI measures:** AQI summarizes daily peaks of the worst pollutant, not long-term exposure to particles or traffic pollution.
- **Missing confounders:** neither dataset includes income, race, smoking, or housing, which strongly affect asthma.
- **Modeled estimates:** PLACES builds county estimates from state-level survey data, which can make counties in the same state look alike and partly explains the large state effect.

4.3 Limitations

- PLACES covers adults aged 18 and older only, so the findings say nothing about childhood asthma.
- Only monitored counties (827) have AQI data, which skews the sample toward larger counties (median population about 120,500, compared with about 26,000 for all U.S. counties).
- The 2026 AQI file was produced mid-year, and many counties show fewer than 100 monitored days. The asthma estimates also come from an earlier survey period.
- Both files are single-year snapshots, so no time trends could be analyzed.
- The analysis is correlational at the county level, so it cannot show causation for individuals.

5. Conclusion and Recommendations

County-level air quality was not associated with adult asthma prevalence across 769 U.S. counties, and the result held across correlation, group comparisons, and regression with and without state effects. County size and state mattered more. This does not show that air pollution is harmless, because county averages, adult-only estimates, and missing socioeconomic data can hide real effects.

I recommend the following next steps:

1. Repeat the analysis with census-tract data in Los Angeles County to capture neighborhood-level exposure.
2. Add income, poverty, and race and ethnicity from American Community Survey data to test for confounding.
3. Study childhood asthma, which may be more sensitive to air pollution, using child-specific data.
4. Use multiple years of air quality data and long-term PM2.5 measures to match the survey period and better represent chronic exposure.
5. Avoid targeting asthma programs using county AQI alone, since high-asthma counties with good air quality exist.

References

- Centers for Disease Control and Prevention. PLACES: County Data (GIS Friendly Format), 2025 release. data.cdc.gov.
- U.S. Environmental Protection Agency. Annual AQI by County, 2026. AirData pre-generated data files.
- Los Angeles County Department of Public Health. Childhood asthma brief.
- Spatial variation of asthma rates in Los Angeles County by environmental and socioeconomic indicators. Environmental Health, 2026.

AI Use Statement

The Introduction and Data and Methodology sections were drafted with AI assistance (Claude) from my analysis notes. The Key Findings, Discussion, and Conclusion were written from my analysis outputs and refined by me. I checked all numbers against the output of my Colab notebook.